

TCP/IP Model-IP Addressing - II

Module 10

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Video: <https://www.youtube.com/watch?v=FQzPOhnEkjg>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

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hello class in the previous lecture we had talked about the flow of data from a application layer to the network layer now we said that from application layer the payload gets transported to transport layer where the segment is formed and then it is transported to the network layer where a packet is formed now this packet is formed with the help of two two things one is called The Source IP address and other is called the destination IP address so we discussed about the various classes of IP addresses and we talked about class A IP addresses Class B and Class C and we have taken one example where we wanted to have a network establishment of a 20 machines so we are going to choose an IP

192.168.1.0 and I want to address all the machines in my network where the wastage would be as less as possible so we are going to utilize cidr which is called classless interdomain routing and this classless Internet Internet domain routing is going to prove sufficient enough in order to lessen the wastage of IP addresses so 192.168.1 1.0 is our Network address and in order to address 20 machines we'll require five bits now how we come to this five bits so you need to memorize a table which is called 2⁰ to 2⁷ because computer is going to use a binary system 2⁰ to 2⁷ now if you convert into the decimal format this is 1 2 4 8 16 32 64 128 and it will go 2⁷ to 2⁸ and so and so forth now these are the eight bits of

the last octet so out of these one 2³ to 2⁵ these five bits we want that that these should address host and these three bits should address the network now wherever the weight is one that will represent Network and wherever the weight is zero that will represent host so now we'll add up these values 128 64 and 32 so this comes out to be 224 that means the last octet in 3 minutes, 1 second our case is 255 do 255 do 255 do 224 now this 224 is going to give us five bits as hosts so our address is 192.168.1.0 is our Network address the first machine will be 192.168.1.1 second will be 192.168.1.2 and it will go up to 192.168.1.20 so these are the 20 machines which will be on our Network now the thing comes what would be the broadcast address of my network now how to calculate the

broadcast address so we are going to write 192.168.0.255

192.168.0.0 and subnet mask is 255.255.255.224 now whenever we have to do some calculations on IP addresses always remember that first of all take the incomplete block so here the incomplete block is 224 so you subtract this from 256 so this comes out to be 32 so the block size is 32 that means in each block there will be 32 machines which can get address from this IP scheme now the last address from this will be the broadcast so if you say 32 - 1 this comes out to be 31 so that means 192.168.0.31 is the broadcast address 192.168.0.1 is the network address from 1 to address 30 this is the wastage because we only required 20 machines so this is how we are going to calculate the number of machines on a network what is the network address what is the broadcast let's take another example

192.168.1.0 this is the network address calculate broadcast address first host address last host address number of hosts and number of subnets if we want to have 10 machines on the network now let's do this so we have been given address 192.168.1.0 now we have to address how many machines 10 machines so let's again draw that table 1 2 4 8 16 32 64 128 so total there are 1 2 3 4 5 6 7 8 bits so these are the eight bits which will address 256 machines so we need to address only 10 machines so in order to address 10 machines I would require four bits so four bits are required for addressing this so that means these four bits will represent Network and these four bits will represent host so 2 to the power of 4 - 2 is the number of

hosts which comes out to be 14 so we'll be able to address 14 machines with help of this addressing mechanism so what would be my subnet mask let's calculate so my subnet mask is 255.255.255.240 so 255.255.255.240 and four zeros so if we convert this into decimal this comes out to be 240 so all the weights which are being given one value you need to add those so 1 2 4 8 16 32 64 128 1 1 1 1 0 0 0 0 so these values are to be added so 128 + 64 + 32 + 16 this comes out to be 240 so this 240 is my last octet so your last OCT dat is 240 so the address will be 192.168.1.0 and the subnet mask will be 255.255.255.240 so that means I am going to consume four bits for host and 28 bits for Network because my IP V4 address is of 32 bits so out of these 32 bits only

four bits are being used as host and rest 28 are being used for Network so we can also specify in another format 192.168.1.0 /28 so this is called cidr notation so IP address can either be specified in a subnet MK mask notation or they can be specified as cidr notation one and same thing so our network is 192.168.1.0 /28 so this 28 is basically 255.255.255.240 now let's calculate the block so block is calculated with the help of incomplete octet so here the fourth OCT is incomplete rest all are complete 255 is the last value which this octet can have so subtract it from 256 so 256 minus 240 this gives me the value 16 so 16 is my block size so the first address in the block is my network address

192.168.1.0 this is the first address and the last address in the block will be my broadcast address which is 192.168.1.5 so these are the addresses for my network so 192.168.1.0 is the network address 192.168.1.5 is the broadcast address first machine will have 192.168.1.1 as the address and the last machine will be 192.168.1.10 because I had only 10 machines in my network but we have few more addresses 112 13 and 14 so these four addresses are going to get faced if we are going to use CIDR notation in order to address 10 machines on the network so these addresses are being used at Network layer so the network layer is going to form a packet this packet will have Source IP address which is the address of my

client machine and this will have destination IP address the address of the server machine and it will have the data from the upper layer transport layer and application layer data so this particular layer is called host to host communication layer so this is going to address a host as a client and host as a server now once this packet is prepared as I told previously also the communication has to happen vertically down so from the network layer the data is going to get transported to the data link layer now what happens at data link layer now at data link layer pdu is called frame so the frames are going to get formed at data link layer and this is hardware dependent layer so that that means if I'm using WiFi then the frame format is different if I'm going to use wired Network the

frame format is different but whether it is a WiFi network or a wired network one thing is very common that we'll have a source Mac address and destination Mac address so these Mac addresses are called physical addresses so IP addresses logical address which is at Network layer and Mac address is physical address which is at data link layer now if you want to uh just correlate this with some analogy so it's very similar that each of you have a mobile phone so your mobile number is basically a logical number like if your mobile number is a 10 digit number so this number will represent a logical number like if you change your uh cell phone and if you want to take a new number the same number can be utilized by somebody else so this is a logical number and then you

have a physical number which is assigned to your mobile set this is called IMEI number so this is your physical address on the same lines on a network we have a physical address and we have a logical address so logical addresses are at Network layer and physical addresses are at data link layer so that means when I want to go to a data link layer I have to form a frame now frame is being formed with the help of source Mac address and destination Mac address now source Mac address or a destination Mac address these Mac addresses are of 48 bit so this is a 48 bit number which is in hexadecimal notation it is defined in a hexadecimal format like 00:00:C0:AB:CD:01 so this is a typical Mac address now this Mac address

has two parts one is called the manufacturer another is the serial number so first three bytes of a MAC address is going to have OEM specification so OEM is original equipment manufacturer so the first three bytes represents OEM so this card can be from Cisco it may be

from Intel it may be from dink so this is the number which will represent that company and after that three bytes they will represent the serial number of the net network card so that means we have a MAC address which basically gets mapped to the IP address which is again going to get ma to the address which is represented in the address bar of a browser so this is complete hierarchy which is being followed from an application layer to data link layer whenever we want to Traverse or whenever we want to surf internet

that means from an application layer we specify a string why we specify string because we as a human being are very good at remembering names we are not good at remembering numbers but as a machine computer system only works on numbers so we need to convert this naming convention into the number convention and vice versa the number convention gets translated into naming convention so that that human and machine can work together so we specify a string like `www do google.com` now the computer doesn't understand what is `www.google.com` so it asks me to get some logical number for this so This logical number is given at Network layer so there will be one IP address which is being given to `www.google.com` now this IP address will

finally get converted into a MAC address which is the address being specified on that particular host so that particular host will have a 48-bit number and that number will represent the MAC address now as an experiment if you want to see what is my IP address and what is my Mac address it's very simple you need to go to the command prompt and at the command prompt you enter a command called `ip config /all` now this is going to yield IP address and Mac addresses for your interface like here in this case if you can see this we have an IP address assigned to this machine 192.168.1.2351 and this is the subnet mask so this is the IP address and subnet 19 minutes, 1 second mask combination on this particular machine as well as if you can see

here this is your physical address 00 5056 01 so this Mac address is the physical address of this particular machine so these addresses are unique so that means this unique address gets translated into ipv4 address which again gets translated into some name like `www.google.com` this is how we can see the IP addresses now let's say I want to know what is IP address of `www.google.com` so you simply type `ping` this is a ping command there is a command whenever you are got connected to the internet you give command `P` followed by `www.google.com` now you see it pops up an address now you would be very familiar about this address this is 173.194.126.80 so this is the IP address which is assigned to `www.google.com` at this point of time so this IP address by virtue of this IP

address I'll be able to make a frame so let's copy this address and make a frame copy now let's create a network packet first then we are going to form a frame so in the network packet we'll have a source IP address which is the IP address of my machine and then we have a destination IP which is the IP which we have got from Google so this is the IP address which is get going to get translated here in the destination field so in the destination field we'll have

this IP address now then this packet goes to the data link layer now what is the job of a data link layer data link layer is going to translate the packet which has come from a network layer and makes a frame out of it now one common mistake which is made

by students is that the frame Mac address source Mac address is the MAC address of the client and destination Mac address is the address of the server which is not actually the case this destination Mac address is the MAC address of your gateway now let's learn this concept that why this destination Mac address is not of the server but this Mac address is of the 22 minutes, 1 second Gateway now as previously we had done let's take the same example 192.168.1.1 let's say this machine wants to talk to 192.168.1.3 machine now the first thing which is being done by your computer is it's going to calculate the network address of both of these addresses if the network address is same in that case we are going to request the

physical address from the Machine by way of broadcast let's see how this communication will happen

192.168.1.1 wants to talk to 192.168.1.3 so that mean this machine knows about the IP address of another machine so they are at which layer they are at Network layer because Source IP is also known and destination IP is also known now in order to make a frame I would require a source Mac address and I would also require a destination Mac address so the source Mac address will be the MAC address add of my machine so I can get it from IP config /all command so that Mac address will go here now what about this Mac address now if the machine is on a same network it is going to request the MAC address of another Machine by way of broadcasting now this broadcasting is called as ARP broadcasting AR RP stands for

24 minutes, 1 second address resolution protocol so if machine knows the IP address of another machine it can only be at Network layer now because it has to go down to the data link layer in order to access the frame so it needs to acquire Mac address and Mac address can be require acquired with with the help of ARP which is called address resolution protocol so my machine is going to send a ARP broadcast and another machine 1.3 in our case is going to reply back with ARP reply which is unicost that means my machine is going to say that what is Mac address of 1.3 then 1.3 is going to say back that my Mac address is 00 0 0 0 C the whole 48 bit number now with the help of this 48-bit number my machine will be able to form a frame and with this Frame I'll be able

to go to the data link layer so that means we'll specify a string that that string is converted into some logical number which is called IP and this IP is going to get converted into a physical address now this is the case when we are on a same network now if we are on a different network this Mac address will be the MAC address of router so this we are going to explain in the next lecture that what is a router for this lecture let's first of all understand what we have learned so far for so that we can recap 26 minutes, 1 second this so in this lecture we understood that classful addressing scheme wastes IPS so we can adapt to cidr

notation now this cidr notation is going to help us to lower

down the wastage of IP addresses so as the number of machines are required those many bits can be chosen from the IP address and then we need to calculate the network address and broadcast address so we learned about how to calculate the network address broadcast address first host address last host address also we found out that after knowing this addresses we need to get a MAC address so that we'll be able to prepare the frame so Mac address would be delivered with the help of ARP protocol which is called address resolution protocol and this Mac address is of 48 bits now out of these 48 bits first three bytes that means 24 bits will represent the OEM which has manufactured this particular card and then next 24 bits they will represent the sequence number

sequence number of the card getting produced this is how the machines are going to talk from an application layer where we are going to specify just a string and then the network layer in between there is a transport layer which is responsible for specifying the process addresses so process addresses were called Port addresses and then we have a network layer which is going to address the IP and finally we have a data link layer which is going to tell us about Mac addresses so each of these layer transport layer is called process to process communication Network layer is called host to host communication and data link layer is called node to node communication so with these Concepts we are ready to

go into the topic which is called security and we learn about how these addresses are helpful in order to get to a particular host and there are certain terms which will come up like we can spoof these IPS and how this spoofing is done that is a part of the next lecture thank you thank you very much